

Arijit De

Kolkata, India

+918981800329 | arijitde2050@gmail.com | <https://github.com/arijitde92> | <https://www.linkedin.com/in/de-arijit/>

PROFESSIONAL OVERVIEW

Machine Learning Engineer with a PhD having 6 years of industry experience spanning computer vision, multi-agent AI systems, and MLOps. Designed and built scalable, robust multi modal RAG pipelines, 3D/2D Image Segmentation and Object Detection solutions across construction, healthcare, and automotive domains. Proficient in Python, FastAPI, Langchain, CrewAI, PyTorch and cloud deployment (GCP, AWS).

WORK EXPERIENCE

Machine Learning Engineer | mVizn, Singapore (Remote) JAN 2024 – PRESENT

- Led development of a multi-agent platform for construction domain (**CrewAI, Python, FastAPI**) powering Autonomous Project Management & Scheduling (APMS), and intelligent project information retrieval — serving 100 concurrent users / 3000 agent runs per day and increasing business efficiency by 50%.
- Hardened multi-agent workflows for reliability by adding failure-mode handling (retries, timeouts, tool-call fallbacks, and output guardrails), reducing task failure rate by 60% and sustaining 99.4% uptime.
- Instrumented end-to-end observability, tracing, and performance monitoring with **CrewAI Tracing** and **DeepEval**, lifting evaluation scores (faithfulness / answer-relevancy) by 75%.
- Built multi-modal Retrieval-Augmented Generation (**RAG**) pipelines over massive text, image, and CAD datasets (10M+ vectors in **Pinecone**) using hybrid search and reranking, achieving retrieval accuracy (recall@k) of 92% while holding retrieval latency under 600 ms.
- Applied **LoRA fine-tuning** to adapt Gemini LLMs for domain understanding and structured output, raising structured-output validity to 98% and cutting token/inference cost per request by 30%.

Machine Learning Engineer | Synerjix, USA (Remote) JUN 2025 – DEC 2025

- Built a Smart Moving Assistant that estimates shifting cost of items from room photos, improving stakeholder business efficiency by 50%, adopted by 3500 users / 10000 estimates processed.
- Deployed DinoV3-based object detection at 65.3 mAP, serving inference at 10ms via **Python + FastAPI, MySQL**, and **AWS** (SageMaker, EC2, EKS) with CI/CD and **Docker**.

Computer Vision Engineer | Institute of Neurosciences, India JAN 2022 – DEC 2023

- Designed and implemented custom 3D Computer Vision solutions for Neurological disorder analysis (**PyTorch, Python, OpenCV, Numpy**), improving diagnosis speed by 40% and reaching 88.68% accuracy on clinical scans.
- Implemented **MLflow**-based experiment tracking, model versioning, model registry, and reproducible training to support scalable **MLOps**, cutting model iteration/turnaround time by 50%.
- Optimized models with **ONNX** and deployed via Docker and **GCP** (Cloud Storage, Compute Engine VMs, IAM, Cloud SQL), reducing inference latency by 65%.

Machine Learning Engineer | Mercedes-Benz Research and Development India, India AUG 2018 – SEP 2019

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection by fine-tuning **YOLOv3** on large-scale annotated datasets, attaining detection mAP50 of 83% at real-time inference speeds.
- Designed & deployed model training, evaluation, and validation workflows in **AWS**.
- Built automated quality assurance (**QA**) and evaluation frameworks using Python, improving annotation throughput by 20% and data quality by 25%.

Systems Engineer | Tata Consultancy Services, India AUG 2014 – APR 2015

- Maintained high-availability backend systems (Java, MySQL) for TCS's internal website, ensuring seamless functionality for over 100,000 users. Streamlined feature release cycle, reducing deployment time by 20%.

EDUCATION

PhD | Jadavpur University, India Graduated: 2024
M.Tech. Computer Science & Engineering | Jadavpur University, India | GPA - 8.83 Graduated: 2018
B.Tech. Computer Science & Engineering | West Bengal University of Technology, India | GPA - 8.81 Graduated: 2014

SKILLS

Languages:	Python, C++, SQL
LLM/Agent Systems:	Multi Agent Orchestration (Crew AI, Google ADK, LangGraph), Agentic Memory, LangChain, LLM, APIs (OpenAI, Gemini, Claude), DeepEval
Frameworks / Libraries:	FastAPI, Flask, PyTorch, Scikit-learn, OpenCV, Open3D, PyTest,
Cloud & Ops:	Google Cloud Platform (GCP), Amazon Web Services (AWS), Git, Docker, MLFlow, Github Actions